DV M-I

1. Perform a practical to install Tableau and explore the Tableau workspace environment & Create basic visualizations using Tableau worksheets.

Steps :- Tableau Installation

Open Browser - tableau.com

Tableau for students free

feel every details

User name password etc.

& download or install

Open Tableau

select sample data set (data source).

Drag → Orders Table —— & connect Returns table

Orders —— Returns

Then click on new sheet I

Scroll the columns.

& save the file

Create hirrarchy

Product ~~ID~~ Name → Right click → click on hirrarchy & Create hirrarchy

Name - Product Name → OK

Make 2 columns

Product Name

- Product Name
- Product ID I you can exchange it both column.

Create visualization.

Drag in columns

Quantity - Sales - Profit

Drag in Rows

Product ID - Product Name -

Create location Hirrarchy

City - Hirrarchy - Create H - Location - OK

location

- Country / Region
- City
- State

2nd visualization → In new worksheet.

Columns Product ID

Rows - Quantity , sales , Profit

& Save the file.

M-1

2. Perform a Practical to connect Tableau with multiple data sources.

Steps :- open tableau

Select sample data set

Drag — Orders — Returns.

Sheet 1 Create hirrarchy

Product ID → create — Product Name — ok

Product Name
- Product ID
- product Name
- category
- Sub category

City/Region → create — location — ok

Location
- City
- country / Region
- State

Sheet 1 rename → calculator column.

~~Ctrl~~ click on arrow [Sum] = ≡ △→ click on arrow

Create calculated field

Name → Profit Ratio

Formula.. sum ([Profit]) / sum ([Sales]) — ok

Create visualization

Drag columns — State Sum Profit (change in average)

Drag columns - Longitude

Rows - Latitude

Marks.

Drag in - country, State

M1

3. Perform a practical to create bar charts & line charts using Tableau.

steps :-

Same as practical 2

only change charts, Graphs.

1) Perform a practical to create symbol maps using geographical datasets.

steps!-

step 1 - Connect to the sample superstore dataset

- The data set connects; the state field appears in the data pane with a globe icone, indicating a recognised Geographic Role.

step 2 Drag state to the View

- Tableau automatically generates Latitude (generated) & longitude (generated) & places them on Rows & Columns, a blank outline map of the country appears.

step 3 Change Mark Type to Circle

- The marks card dropdown is changed from automatic to circle, a small circular mark appears at the centroid of every state.

step 4:- Add sales to size

sum (sales) is dragged onto the size shelf of the marks card, circles now scale in proportion to each states total sales.

step 5:- Add sales to color and add a title

- sum (sales) is also added to color, producing a size-&-colour encoded symbol map, the sheet is titled 'Sales by state'.

2) Perform a practical to design interactive dashboards using filters.

steps :-

step 1 - Create the individual worksheets
- Two worksheets are built from sample-superstore 'Sales by category' (bar chart) & 'Sales by Region' (bar chart)

step 2 - Create a New dashboard
- A new dashboard tab is created, a blank canvas opens with size, layout & objects options in the left pane

step 3 - Drag worksheets onto the dashboard
- Both worksheets are dragged from the left pane onto the canvas & arranged side by side using the tiled layout.

step 4 - Add a Filter Card
- The Category field is dragged onto the dashboard as a filter, a filter control listing Furniture, Office Supplies & Technology appears.

step 5 - Apply the filter to All worksheets
- The filters dropdown menu is set to 'apply to worksheets' - All using this data source, so one selection updates every chart at once.

step 6 Format and Title the Dashboard
- The dashboard is titled "Sales overview Dashboard" & minor formatting is applied for a polished look.

3] Perform a Practical to create density maps for spatial analysis.

Step 1 Build a Point Map of order locations.

- City with Latitude / Longitude generated is placed on the view, a point for every order location appears, density overlapping in large metro areas.

Step 2 Change mark Type to Density.

- The marks card drop down is changed density, overlapping points are blended into a continuous heat-map surface.

Step 3 Adjust the colour palette / Intensity

- The density colour scheme is changed to a blue → Yellow → Red palette, intensity settings are adjusted so hotspots stand out clearly.

Step 4 - Add numbers of records to control weighting

- The number of record field is added to the marks card so that areas with more individual orders glow more intensly on the density surface.

step 5 Add a Title to legend

- The sheet is titled "Order Density Map" & the intensity legend is displayed to help interpret the ~~colo~~ colour scale.